

Assignment – 7

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Task 7.1

Create a project with different user groups and implement group policies.

Repos → Branches → master → Branch Policies:

```
az repos policy create --repository-id <repo_id> \
  --branch master \
  --blocking true \
  --enabled true \
  --policy-type Build \
  --settings '{"buildDefinitionId":<build_def_id>,"queueOnSourceUpdateOnly":true,"manualQueueOnly":false,"displayName":"Require Successful
```

Or Set group permissions on repository and branches for existing group

```
1 # Contributors: no direct push
2 az repos permission update --repository-id <repo_id> \
3   --branch master \
4   --user-group "Contributors" \
5   --deny "ForcePush,Push"
6
7 # Project Administrators: full access
8 az repos permission update --repository-id <repo_id> \
9   --branch master \
10  --user-group "Project Administrators" \
11  --allow "ForcePush,Push,CreateBranch"
```

Task 7.2

Apply branch policies such that only the project administrator can access the master branch; contributors cannot.

In Repos -> Branches -> master -> Branch Policies

Policy Settings:

- ✓ Require a minimum number of reviewers: 1 or more
- ✓ Check for linked work items
- ✓ Limit merge types (disable "direct push")
- ✓ Automatically include code reviewers
- ✓ Use status checks from external services (e.g., build pipelines)
- ✓ Enforce a successful build
- ✓ Require comments resolution

Disable direct pushes:

In Project Settings -> Repositories -> master, set:

Contribute = Deny for Contributors

Contribute via pull request = Allow

```
az repos policy required-reviewers create \  
  --repository-id <repo_id> \  
  --branch master \  
  --blocking true \  
  --enabled true \  
  --settings '{"requiredReviewerIds":["<admin_user_id>"]}'
```

Task 7.3

Apply branch security and locks.

```
# Lock the master branch (optional)  
az repos ref lock --name "refs/heads/master" --repository-id <repo_id>
```

Task 7.4

Apply branch filters and path filters.

In Build Pipeline Trigger:

```
✓ trigger:  
✓   branches:  
✓     include:  
✓       - feature/*  
  
✓   paths:  
✓     include:  
✓       - src/**  
✓     exclude:  
✓       - docs/**
```

Task 7.5

Apply a pull request.

Admin/Reviewer must approve and complete PR.

```
az repos pr create \  
  --repository <repo_name> \  
  --source-branch feature/login-fix \  
  --target-branch master \  
  --title "Fix login issue" \  
  --description "Fixes the bug with login"
```

Task 7.6

Apply branch policy, Apply branch security.

```

1 # Build validation policy
2 az repos policy build create \
3   --repository-id <repo_id> \
4   --branch master \
5   --blocking true \
6   --enabled true \
7   --settings '{"buildDefinitionId":<build_def_id>, "queueOnSourceUpdateOnly":true, "manualQueueOnly":false, "displayName":"Require Success
8

```

```

1 # Work item linking policy
2 az repos policy work-item-linking create \
3   --repository-id <repo_id> \
4   --branch master \
5   --blocking true \
6   --enabled true
7

```

Task 7.7

Apply triggers in build and release.

Go to Pipelines -> Releases -> Your Pipeline -> Pre-deployment conditions -> Gates

Example Gates:

- *Invoke REST API*
- *Query Azure Monitor Alerts*
- *Check work items or approvals*

```

✓ trigger:
✓ branches:
✓ include:
  - feature/*

```

Task 7.8

Apply gates to the pipeline.

Pipelines → Releases → Environments → Pre-deployment conditions → Gates

Enable and configure:

- *Query Work Items*
- *Invoke REST APIs*
- *Azure Monitor alerts*

```

✓ environments:
✓ - environment: 'prod'
✓ checks:
✓   - task: InvokeRESTAPI@1
✓     inputs:
✓       url: 'https://your-check-endpoint.com'
✓       method: 'GET'

```

Task 7.9

Apply security on branches so that contributors can create a pull request but cannot directly merge code to master.

```
# Block contributors from merging/pushing
az repos permission update --repository-id <repo_id> \
  --branch master \
  --user-group "Contributors" \
  --deny "ForcePush,Push"

# Allow pull requests
# No extra config required; pull request is still allowed
```

Task 7.10

Use work items in pipelines.

```
az repos pr work-item add \
  --id <pr_id> \
  --work-items <work_item_id>
```

```
- script: echo "##vso[build.addbuildtag]LinkedWorkItems"
  displayName: 'Tag work items'
```